



Data Preprocessing in Machine Learning: Cleaning, Scaling, and Leakage

Preprocessing · Machine Learning · R Language



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What is Data Preprocessing?

Before any analysis, data must be prepared. Preprocessing transforms raw data into reliable information — cleaning inconsistencies, integrating distinct sources, and structuring attributes so that mining and machine learning algorithms can operate with maximum effectiveness.



Data Cleaning

Handles missing values, smooths noise, removes outliers, and resolves inconsistencies to ensure data quality.



Data Integration

Consolidates multiple sources — databases, cubes, or files — into a cohesive base ready for analysis.



Data Reduction

Reduces dimensionality and numerosity of data, optimizing processing without losing relevant information.



Data Transformation

Applies normalization, discretization, and concept hierarchies to adapt data to analysis algorithms.

📖 **References:** Han, Pei & Tong (2022). Data Mining: Concepts and Techniques, 4th ed. • García, Luengo & Herrera (2014). Data Preprocessing in Data Mining.

Data Quality: Why Preprocess?

To preprocess data effectively, one must first understand what makes it problematic. Data quality is multidimensional — failures in any of these dimensions compromise analyses and lead to misguided decisions. Understanding them is the starting point for applying the correct preparation techniques.



Accuracy

Do the data correctly reflect reality? Inaccurate values distort any model or analysis.



Completeness

Are all required attributes and records present? Missing values must be addressed before analysis.



Consistency

Are the data uniform across sources and records? Internal contradictions invalidate comparisons and aggregations.



Timeliness

Is the information up to date? Outdated data may not represent the actual state of the system.



Believability

Is the source trustworthy and the collection process robust? The credibility of the data underpins the validity of conclusions.



Interpretability

Are the data understandable? Adequate documentation and metadata facilitate the correct use of information.

 **References:** Han, Pei & Tong (2022). Data Mining: Concepts and Techniques, 4th ed. • García, Luengo & Herrera (2014). Data Preprocessing in Data Mining.

Transformations

Once the dimensions that define data quality are understood, the next step is to act on them. The `daltoolbox` package organizes this work into eight steps that form a cohesive pipeline: starting from sampling and cleaning — which prepare the foundation — through normalizations, encodings, and transformations that make data ready for modeling, up to dimensionality reduction. The balancing and attribute selection steps depend on the concept of class and therefore appear after the initial variable transformations.

01

Sampling

Train/test split (random and stratified)

03

Normalization

Normalization of numerical values (min-max and z-score)

05

Smoothing

Discretization of continuous variables (interval, frequency, and clustering)

07

Attribute Selection

Identification of relevant predictors (info gain, relief, stepwise, lasso, FSS)

02

Cleaning

Removal of NAs and outlier detection (boxplot and Gaussian)

04

Encoding

Transformation of categorical variables (`categ_mapping`)

06

Balancing

Correction of imbalanced classes (oversampling and undersampling)

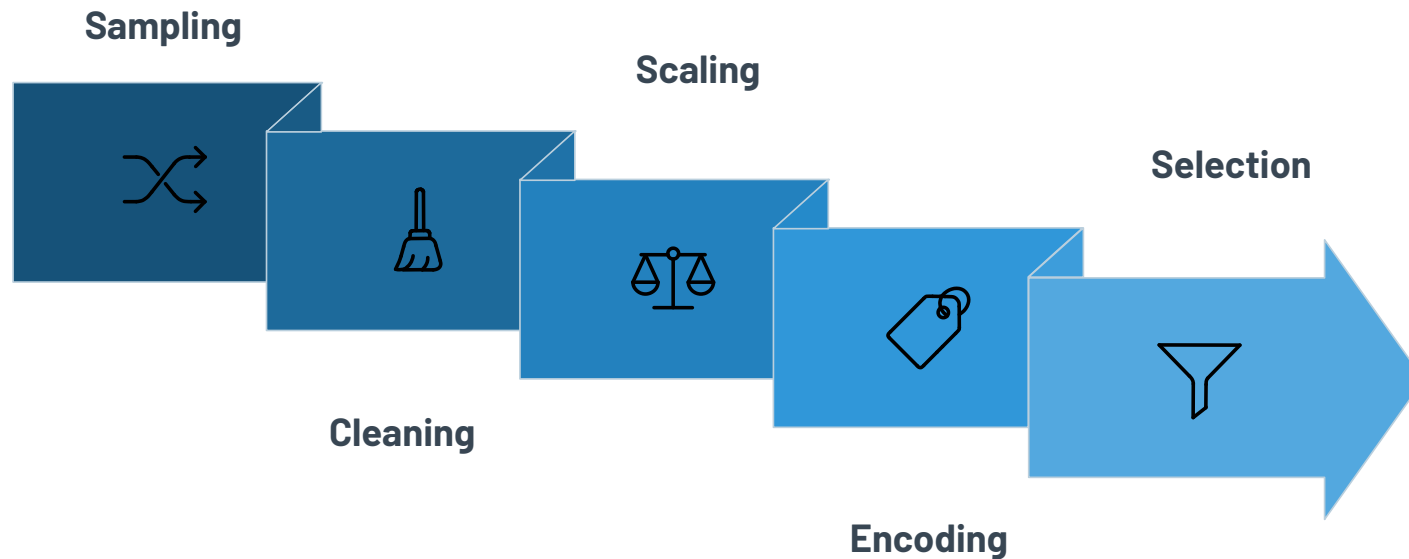
08

Dimensionality Reduction

Compression of the feature space (PCA and curvature heuristics)

Recommended Preprocessing Workflow

Of the eight steps presented earlier, five form the backbone of supervised preprocessing. The diagram below summarizes the recommended order of execution — each arrow indicates a dependency: the next step only makes sense after the previous one is complete.



Following this workflow ensures that each transformation is applied in the correct order, preventing data leakage and ensuring that the model is evaluated under realistic conditions. The next slides detail each of these steps — Sampling, Cleaning, Scaling, Encoding, and Selection — exactly in the sequence presented above.

Summary of Transformations

The flow presented earlier highlighted the five central steps of supervised preprocessing. Expanding that view, the table below consolidates **all eight transformations** covered throughout this material — from the most fundamental to the most specialized. Together, they form a complete repertoire for preparing data before model training, covering everything from the initial dataset split to dimensionality reduction.

Sampling Random or class-stratified train/test split. - sample_random - sample_stratified	Cleaning Removes NAs and detects outliers via IQR or Gaussian distribution. - na.omit - outliers_boxplot - outliers_gaussian	Scaling Normalizes values to [0,1] or standardizes to mean 0 and standard deviation 1. - minmax - zscore
Encoding One-hot encoding of categorical variables. categ_mapping	Smoothing Discretizes continuous variables by interval, frequency, or clustering. - smoothing_inter - smoothing_freq - smoothing_cluster	Balancing Corrects class imbalance through over- or undersampling. - bal_oversampling - bal_subsampling
Feature Selection Ranks or selects relevant predictors using the class as the target. - feature_selection_info_gain - feature_selection_relief - feature_selection_stepwise - feature_selection_lasso - feature_selection_fss		Dimensionality Reduction Projects data onto principal components; curvature heuristics determine the optimal number of components. - dt_pca - fit_curvature_min - fit_curvature_max

 **Package:** All functions are available in daltoolbox. Install with install.packages("daltoolbox"). Optional dependencies: glmnet (Lasso) and leaps (FSS).

References & Additional Sources

Online Resources

- 
- [CRAN Package — daltoolbox](#)
Official documentation for all preprocessing functions.
 - [DAL Toolbox — GitHub](#)
Source code, vignettes, and pipeline examples.
 - [DAL Toolbox — Issues & Contributions](#)
Report issues or contribute to the toolkit.

Key Academic References

- Ogasawara et al. (2026). daltoolbox: Leveraging Experiment Lines to Data Analytics. R package v1.3.767. CRAN.
- Ogasawara et al. (2009). Experiment Line: Software Reuse in Scientific Workflows. SSDBM 2009, LNCS 5566, pp. 264–272. Springer.
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- Hastie, T., Tibshirani, R. & Friedman, J. (2009). The Elements of Statistical Learning. 2nd ed. Springer.